

Heart Disease Prediction Using Logistic Regression

A Comprehensive Analysis Report

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1. Introduction

CVDs are considered a major cause of mortality across the world. This is especially important for CVDs as an early detection will allow for better prevention and frequent interventions. In this project, logistic regression is used to timely input clinical information to reach a comprehensive judgment to early predict individuals likely to have heart diseases. The first aim is to develop a model that agrees with actual values and produces results which the healthcare professionals, may use for decision making.

2. Scope of the Analysis

In this project, a logistic regression model is built from UCI Heart Disease Dataset. The scope includes:

- Preparing the dataset using ETL (Extract, Transform, Load) processes.
- Exploring relationships between clinical features and the target variable (presence of heart disease).
- Training, evaluating, and interpreting the logistic regression model.
- Providing key insights into significant predictors of heart disease.
- Assessing the model's performance using metrics like accuracy, precision, recall, and AUC-ROC.

3. Existing System

Today, the confirmed methods of diagnosing heart diseases involve physical assessment, chest x-rays/ultrasound, blood tests, and others, which are expensive and take a lot of time. Other practical models for diagnosis include predictive models that include logistic regression models that match the traditional diagnostic systems. These models can then effectively and cheaply screen candidates for high risk based on historical data analysis.

4. ETL Process

Explain the Extract, Transform, Load (ETL) process:

- **Extract:** The UCI Heart Disease Dataset was acquired. It contains 303 patient records and 14 features, including age, sex, cholesterol levels, and target (presence of heart disease).
- **Transform:** Discuss cleaning and preprocessing steps.
 - Missing values were checked, and none were found in this dataset.
 - Categorical variables (`cp`, `restecg`, `slope`, `thal`, and `ca`) were encoded using one-hot encoding.
 - Numerical variables (`age`, `trestbps`, `chol`, `thalach`, and `oldpeak`) were scaled using `StandardScaler` to normalize the data.
- **Load:** The transformed data was loaded into the modeling pipeline for further analysis and modeling.

age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope
63	1	3	145	233	1	0	150	0	2.3	0
37	1	2	130	250	0	1	187	0	3.5	0
41	0	1	130	204	0	0	172	0	1.4	2
56	1	1	120	236	0	1	178	0	0.8	2
57	0	0	120	354	0	1	163	1	0.6	2

Table 4.1: Heart Disease Dataset

5. Analysis on Dataset

5.1 Correlation Heatmap

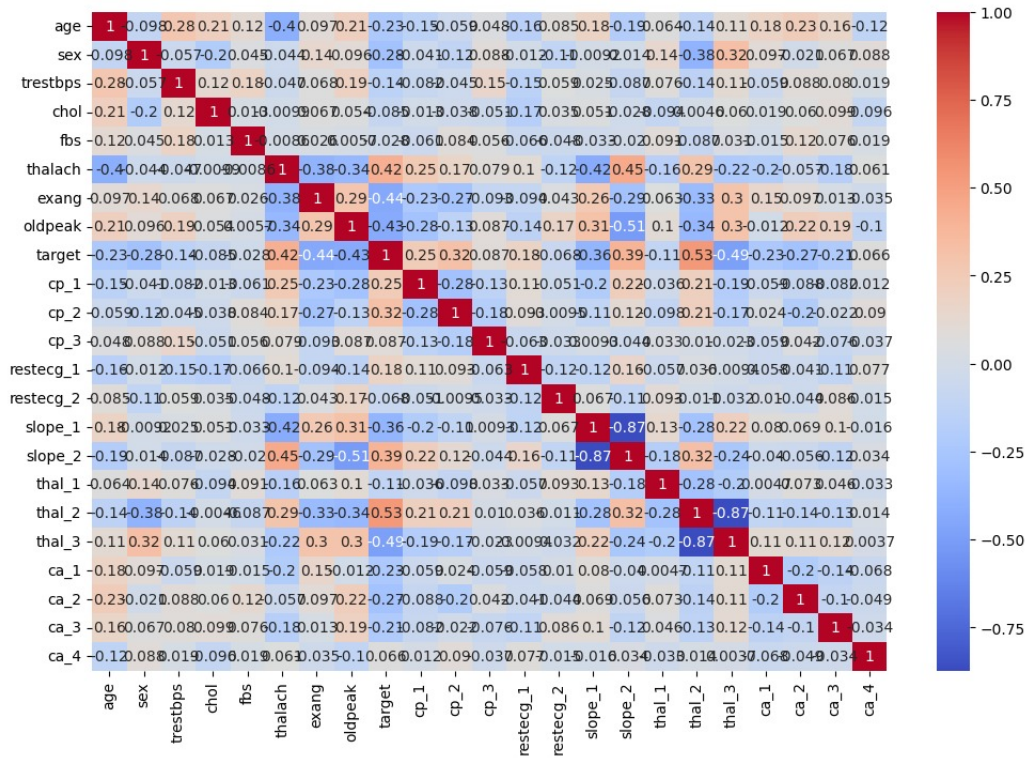


Figure 5.1: "Correlation Heatmap of Clinical Features"

For the comparison of the features with the target variable, a correlation heatmap was created. It was found that the features such as thalach and ca were a significant fit with the target variable.

5.2 Target Variable Distribution

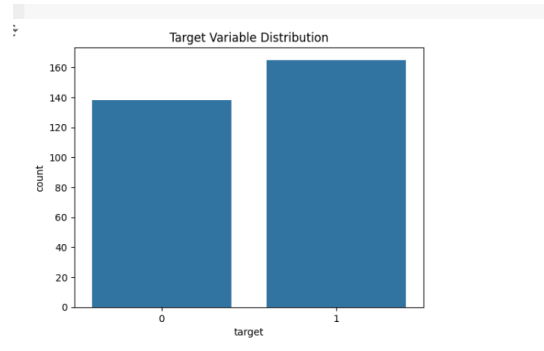


Figure 5.2: "Distribution of Target Variable (Presence of Heart Disease)"

The dataset contains slightly imbalanced classes, as detailed below:

- **Presence of Heart Disease (1):** Approximately 54%
- **Absence of Heart Disease (0):** Approximately 46%

5.3 Pairwise Feature Relationships

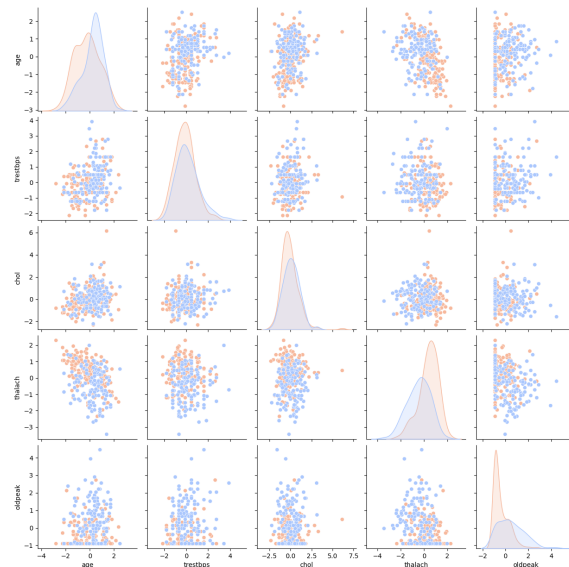


Figure 5.3: "Pairplot for Key Clinical Features by Target Variable"

- Pairplots were created to visualize the relationships between features (age, trestbps, chol, thalach, oldpeak) and the target variable.
- KDE (Kernel Density Estimate) plots on the diagonal showed clear separations in distributions for thalach and oldpeak.

5.4 Categorical Feature Distributions

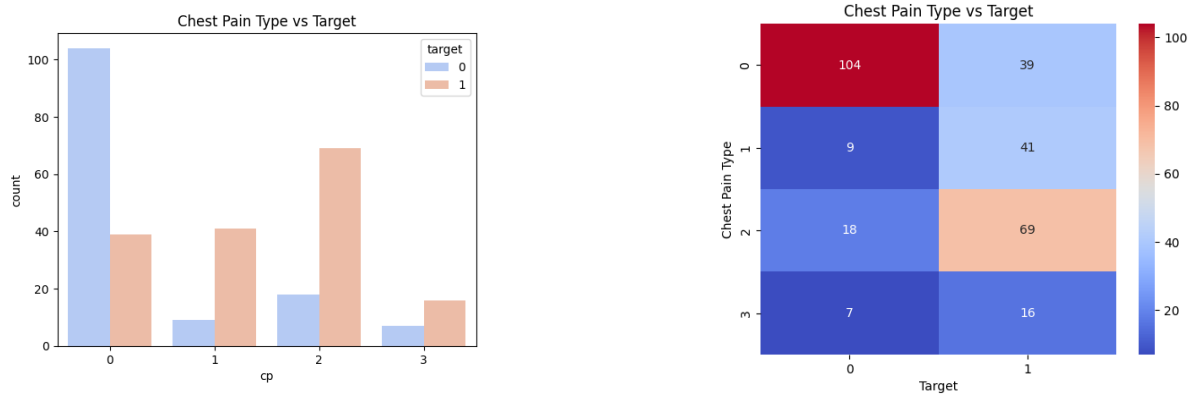


Figure 5.4: "Chest Pain Type (cp) Distribution by Target Variable".

Both count plots underlined the dependency of the target variable on categorical attributes such as chest pain type (cp). Some of them were more linked with the heart disease themselves.

5.5 Numerical Feature Distributions

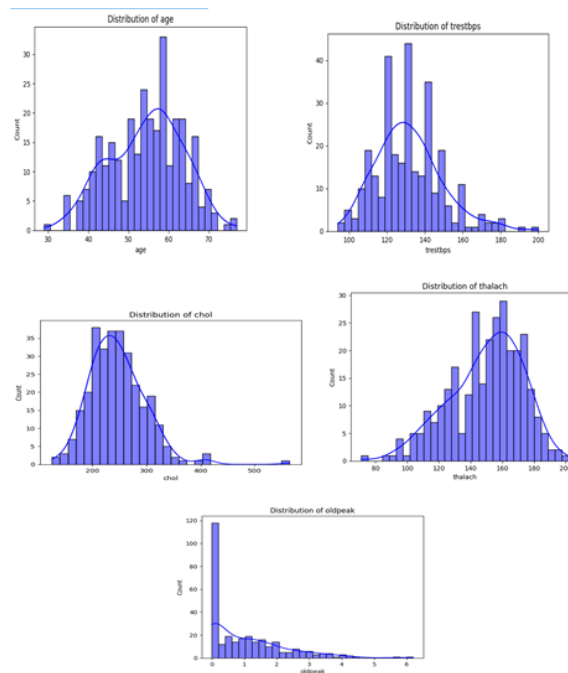


Figure 5.5: "Distribution of Numerical Features"

- Histograms and density plots showed distributions of age, trestbps, chol, thalach, and oldpeak.
- Patients with heart disease generally had higher oldpeak values and lower thalach values.

5.6 Outlier Analysis

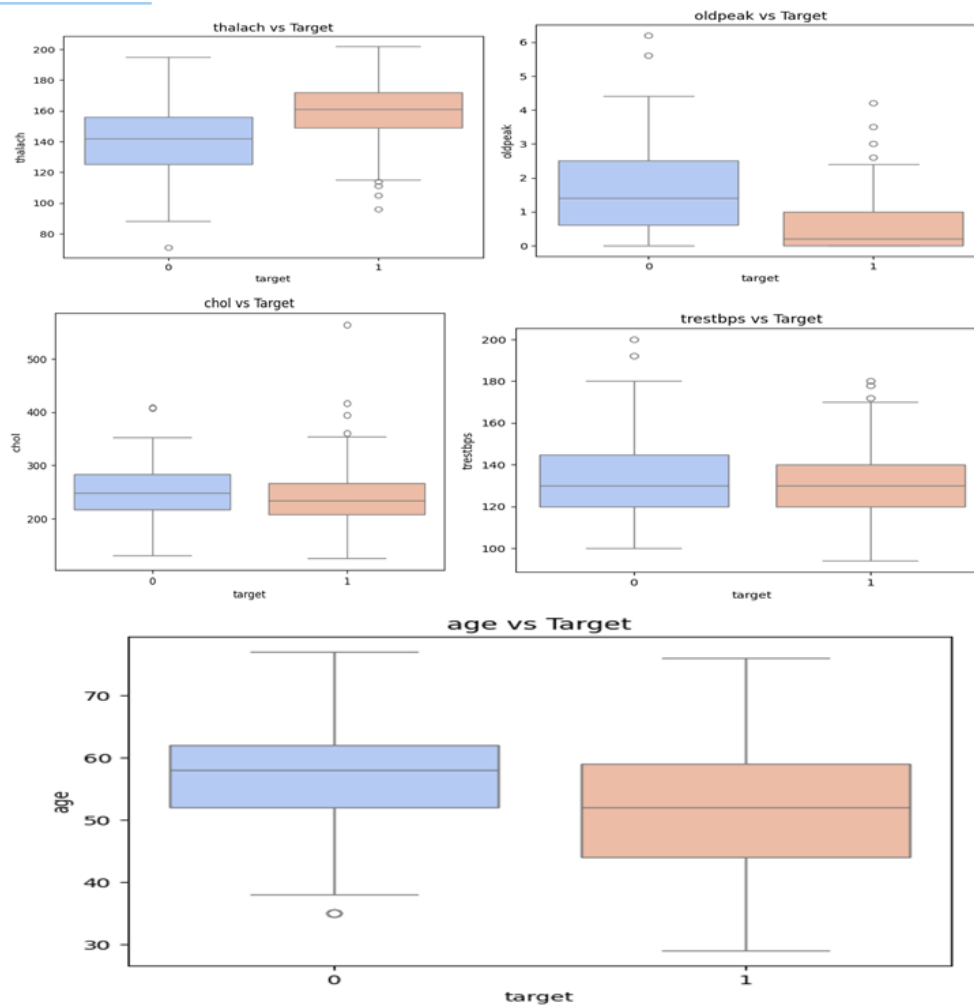


Figure 5.6: "Boxplots of Numerical Features by Target Variable"

- Boxplots were used to detect outliers for numerical features like chol and trestbps. Patients with heart disease had higher medians for chol.

5.7 KDE Plots

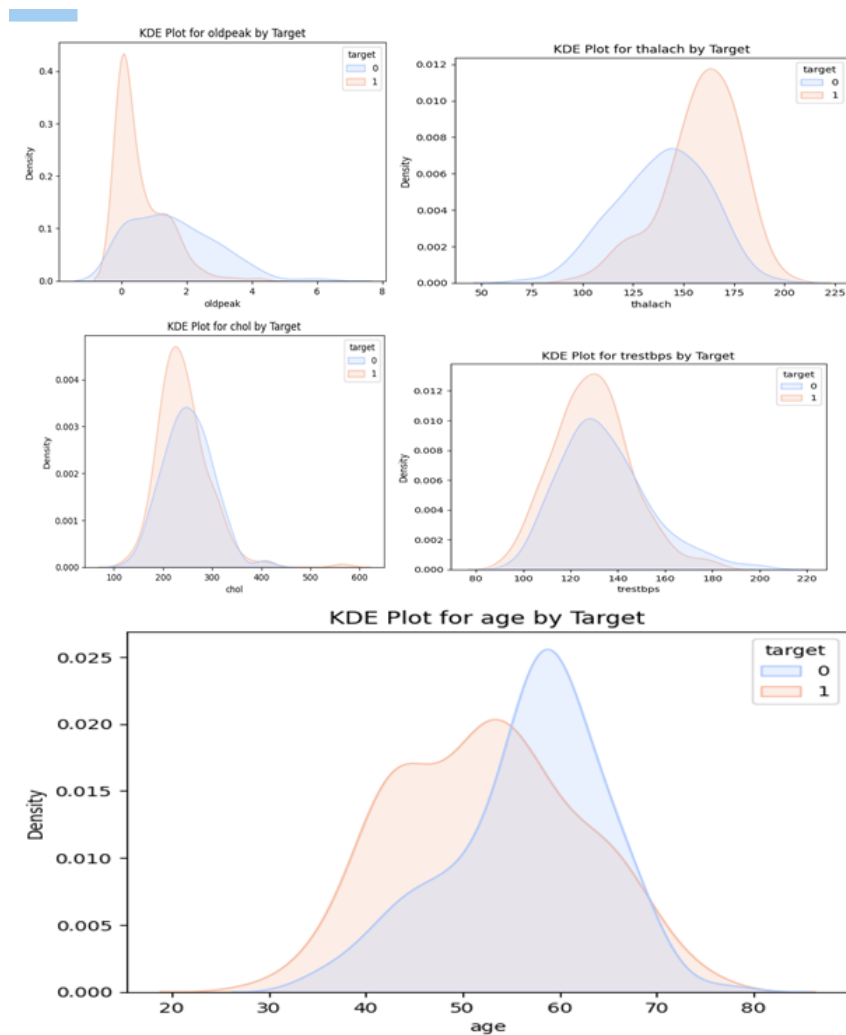


Figure 5.7: "KDE Plots of Numerical Features by Target Variable"

- These plots visualize the distribution of numerical features like age, trestbps, chol, thalach, and oldpeak concerning the target variable.

Description:

1. The KDE plots show how these features vary between patients with and without heart disease.
2. Features such as thalach and oldpeak exhibit clear separation in distributions, indicating their significance in predicting heart disease.

6. List of Analysis with Results

6.1 Logistic Regression Model Performance

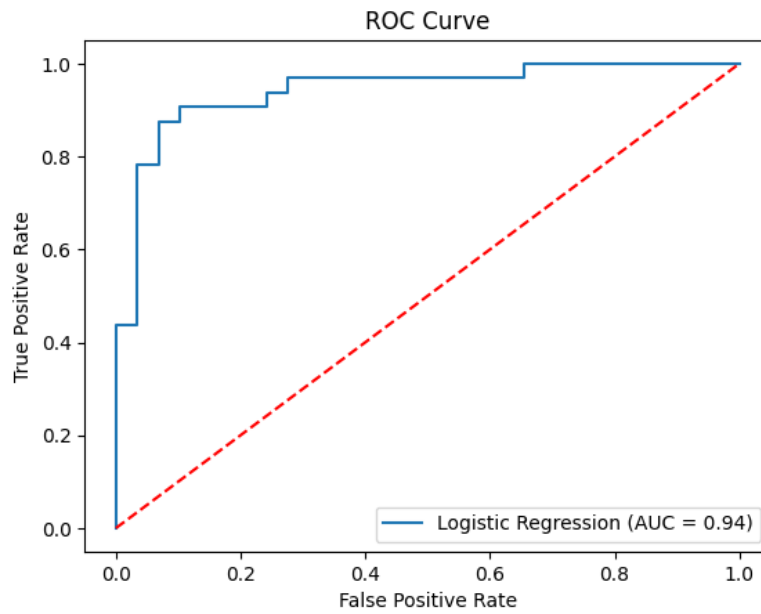


Figure 6.1: "ROC Curve for Logistic Regression Model"

Metric	Value
Accuracy	90.16%
Precision	93.33%
Recall	87.50%
F1 Score	90.32%
ROC-AUC Score	94.18%

Table 6.1: Performance metrics of the logistic regression model.

6.2 Feature Importance

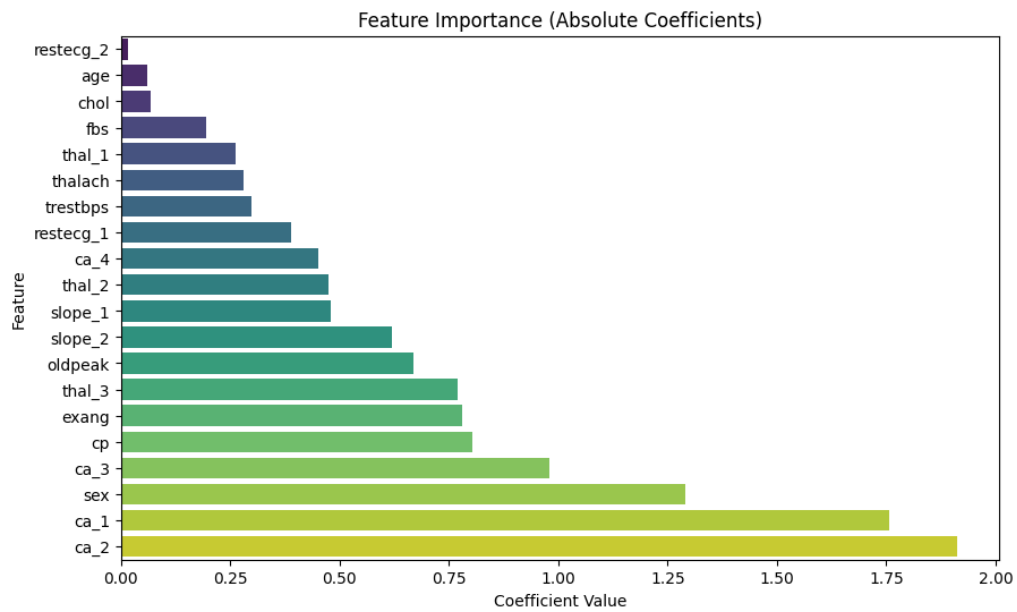


Figure 6.2: "Bar Plot of Feature Importance Based on Coefficients"

Feature	Coefficient	Correlation with Target
ca_2	-1.85	Negative
cp_3	+1.34	Positive
thalach	+0.33	Positive

Table 6.2: Most influential features and their correlation with the target variable.

7. Future Scope

The analysis can be extended in the following directions:

1. **Enhanced Models:** Extend the analysis to include more advanced machine learning models, such as Random Forest, XGBoost, or Neural Networks.
2. **Feature Engineering:** Include additional clinical or lifestyle-related features for better predictions.
3. **Real-World Integration:** Deploy the model as a user-friendly web or mobile application for clinical use.
4. **Longitudinal Data:** Incorporate time-series patient data to track progression and improve predictions.

8. Conclusion

The project also effectively outline and estimate a logistic regression algorithm using the UCI Heart Disease Dataset. By process and data preprocessing, EDA, and feature selection, it was possible to establish the potential predictors like thalach, ca-2 and oldpeak. The accuracy level of 90.16% of the test data, precise 93.33%, recall 87.50%, and ROC AUC 94.18% indicated that model has a very good ability to classify patient with and without heart disease. Correlation heatmaps and KDE plots which were used to analyze inter-feature correlations and distribution informativeness, as well as ROC curves, which showed the efficiency of the models used, were critical for learning purposes. Hence, this project demonstrates the effectiveness of logistic regression in clinical prediction problem and its applicability in practice. Prospective functions include expanding features, the use of state-of-the-art machine learning, and integration of real-time data can prospectively enhance usefulness in the early identification of cardiovascular diseases and decreasing the global burden of these diseases.

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